High-speed SWIR Hyperspectral Imaging For Material Classification In Operating Room

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***Abstract*—Intraoperative tissue classification requires fast and reliable imaging tools for real-time surgical guidance. Although hyperspectral imaging provides rich spectral information, most existing systems rely on spectral or spatial scanning and are too slow for clinical use. To address this limitation, we present a high-speed short-wave-infrared (SWIR) hyperspectral projector based on structured illumination and compressive sensing. Spectral information is encoded at the illumination stage using a digital micromirror device, and the resulting measurements are decoded to recover spectral features for tissue classification. By shifting spectral multiplexing to the projection stage, the proposed framework enables faster acquisition and provides a foundation for real-time SWIR hyperspectral imaging.**

***Index Terms*—SWIR, Hyperspectral Imaging, Material Classification, Compressive Sensing, Computer Vision.**

I. Introduction

Intraoperative tissue assessment plays a critical role in many surgical procedures, where fast and reliable imaging tools are required to support real-time clinical decision-making. For example, accurate evaluation of burn severity [1], cutaneous inflammation [2], or identification of diseased tissue margins directly influences surgical outcomes and treatment planning. Conventional visual inspection is often subjective and may fail to capture subtle physiological differences across tissue types. These limitations motivate the development of quantitative optical imaging techniques capable of providing rapid, non-contact, and information-rich measurements during surgery.

Hyperspectral imaging (HSI) has emerged as a promising modality for biomedical applications [3], [4] due to its ability to capture detailed spectral signatures associated with tissue composition. In particular, imaging in the short-wave infrared (SWIR) spectral range offers several advantages over visible and near-infrared regimes. Reduced scattering in biological tissue enables improved penetration depth and more stable reflectance measurements, while stronger absorption features related to water and lipid content provide enhanced sensitivity to physiological variations [5]–[7]. These properties make SWIR hyperspectral imaging attractive for intraoperative tissue classification tasks.

Despite its potential, the clinical translation of hyperspectral imaging remains limited by acquisition speed. Most existing HSI systems rely on spectral or spatial scanning mechanisms [8]–[10], such as pushbroom imaging, tunable filters, or filter-wheel-based approaches. While snapshot hyperspectral cameras can partially alleviate scanning requirements, they often suffer from trade-offs between spatial resolution, spectral resolution, and signal-to-noise ratio. As a result, many current systems are not compatible with real-time surgical workflows, where rapid data acquisition and processing are essential to minimize motion artifacts and workflow disruption.

Compressive sensing has emerged as an effective approach for accelerating hyperspectral imaging by reducing the number of measurements required for spectral recovery [11]. Compared with conventional SWIR hyperspectral systems that rely on costly detector arrays and sequential spectral scanning, compressive hyperspectral imaging enables faster acquisition with simpler and lower-cost hardware. In our previous work [12], we demonstrated that hyperspectral structured illumination based on a digital micromirror device (DMD) and a grayscale camera can recover spectral responses of different materials while also supporting 3D reconstruction. The high switching speed of the DMD indicates strong potential for high-speed hyperspectral acquisition without requiring expensive SWIR focal-plane arrays. Building on this concept, the present work extends this framework toward SWIR hyperspectral imaging for tissue characterization and classification.

In this work, we present a high-speed SWIR hyperspectral imaging framework based on spectral multiplexing at the illumination stage. A digitally controlled hyperspectral projector generates structured patterns that encode spectral information before interacting with the target scene. Each projected pattern produces a compressed measurement captured by a grayscale camera. A computational decoding pipeline then reconstructs spectral features and maps them to tissue classification outcomes. By shifting multiplexing to the illumination domain, the proposed approach avoids conventional spectral scanning and enables faster hyperspectral acquisition for intraoperative use.

II. Method

A. System Overview

The overall workflow of the proposed imaging system is illustrated in Fig. 1. A hyperspectral projector illuminates the target with spectrally encoded structured light, and a grayscale camera captures the reflected responses. The recorded image sequence is then processed to recover the spectra of user-defined regions of interest (ROIs) corresponding to different materials or tissue types. The reconstructed spectra are used as inputs to a neural-network-based classifier, which assigns each ROI to a predefined tissue class. The classification result is then rendered and projected back onto the target scene,

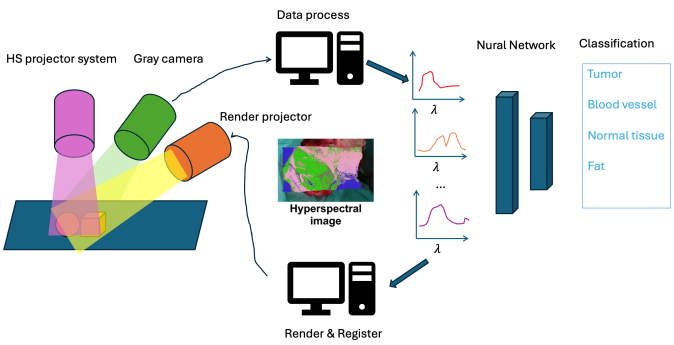

Fig. 1. Overall workflow of the proposed real-time SWIR hyperspectral imaging and projection system. A hyperspectral projector illuminates the target scene, a grayscale camera captures the reflected responses, spectral signatures from regions of interest are reconstructed and classified, and the classification results are rendered back onto the object for real-time surgical guidance.

enabling real-time visual feedback for surgical guidance. The hyperspectral projection module is described in the following subsection.

B. Spectral Encoding Strategy

Fig. 2 shows the optical design of the hyperspectral projector (HSP). Light from a halogen lamp is delivered through a gooseneck fiber-optic light guide and focused by a cylindrical lens onto a vertical slit of width 200 μm. The light is collimated by lens L1 and spectrally dispersed along the y-direction by a transmission diffraction grating (Thorlabs, Visible Transmission Grating, 300 grooves/mm). An achromatic lens L2 focuses the dispersed spectrum onto a digital micromirror device (DMD) (Texas Instruments, DLP6500 0.45 WXGA DMD, 1920×1080). The distances from the grating to L2 and from L2 to the DMD are both equal to the focal length f of L2, mapping different wavelengths to different DMD positions.

Programmable binary patterns on the DMD perform spectral and spatial modulation. The modulated light is reflected back through L2, recombined by the diffraction grating, and redirected by the beam splitter. A cylindrical lens forms a spectrally encoded stripe at the target plane, while a relay-based magnification lens module adjusts the projection size and a rotating mirror controls the stripe orientation.

C. Compressive Spectral Reconstruction

The compressive spectral reconstruction process is illustrated in Fig. 3. Let $X \in \mathbb{R}^{N\times(h\times w)}$ denote the unknown spectral image, where N is the number of spectral channels and each column corresponds to the spectrum of one spatial pixel. The measurement matrix is denoted by $Y \in \mathbb{R}^{N\times(h\times w)}$, where each row represents the response acquired under one projected pattern. The coding matrix $H \in \mathbb{R}^{N\times N}$ is formed by Hadamard patterns, with each row corresponding to one spectral coding pattern. The forward model can therefore be written as

$$Y = HX.$$

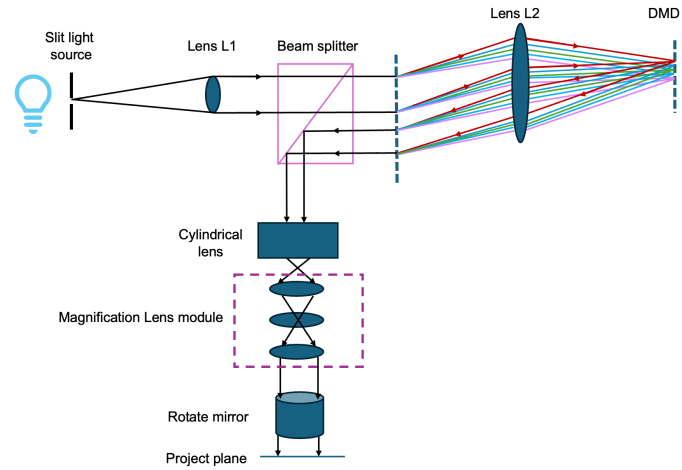

Fig. 2. Optical design of the hyperspectral projector. Broadband light is spectrally dispersed and encoded by the DMD, then recombined and projected as a spectrally modulated stripe. A magnification lens module adjusts the projection size, and a rotating mirror controls the stripe orientation.

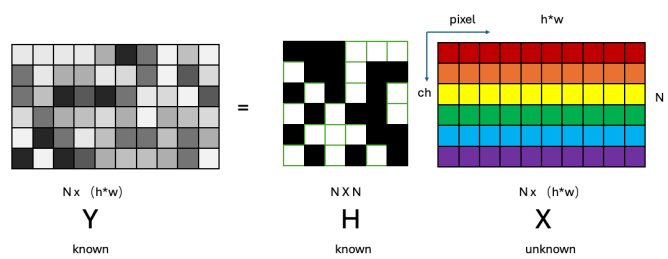

Fig. 3. Compressive spectral reconstruction model based on Hadamard coding, where $Y = HX$.

Since H is full-rank, the spectral image X can be directly recovered by matrix inversion:

$$X = H^{-1}Y.$$

In this work, a 24×24 Hadamard matrix is used for spectral encoding and reconstruction. For physical implementation, the first Hadamard row, which consists of all ones, is divided into two complementary half-and-half measurements to maintain balanced projection energy. These two measurements are added together during data processing to recover the equivalent all-one pattern response. After this preprocessing step, the full measurement matrix is assembled and used for spectral reconstruction.

Each element of Y corresponds to the pixel intensity recorded by the grayscale camera under one Hadamard illumination pattern. Therefore, the measurement matrix Y is formed by vectorizing the captured image sequence, where each row represents the vectorized image acquired at one spectral coding pattern.

After solving the linear model to obtain X, the reconstructed spectral data are reshaped back to the original spatial dimensions of the captured images. In this representation, each spectral channel corresponds to a reconstructed image of size $h \times w$. Regions of interest (ROIs) are first selected in the

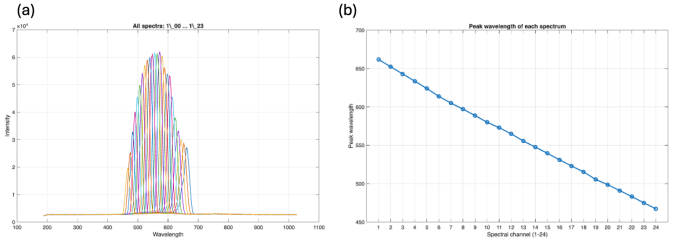

Fig. 4. Spectral channel calibration results. (a) Measured spectra of the 24 spectral channels. (b) Peak wavelength of each channel, showing an approximately linear relationship with channel index.

measurement image domain, and the corresponding spatial locations are then used to extract the same regions across all reconstructed spectral channels.

To improve robustness against noise and local intensity variations, a median filter is applied within each ROI for every spectral channel. The filtered intensity values from all channels are then combined to form the final spectral signature associated with the selected material or tissue region.

III. Results

A. Spectral Channel Calibration

To characterize the spectral encoding performance of the hyperspectral projector, a spectral channel calibration experiment was conducted. The dispersion direction on the DMD was divided into 24 equally spaced regions along the y-axis. A raster scanning strategy was employed to sequentially project narrow spectral bands corresponding to each region, and the resulting spectra were measured.

Fig. 4(a) shows the measured spectra of all 24 channels. The combined result demonstrates that the system provides continuous spectral coverage from 450 nm to 680 nm with approximately uniform channel spacing. This confirms that the DMD-based spectral encoding enables controlled selection of wavelength bands.

Fig. 4(b) presents the peak wavelength extracted from each measured spectrum as a function of channel index. The peak wavelength exhibits an approximately linear relationship with channel number, indicating that the spectral dispersion mapping onto the DMD is well calibrated. This linear trend verifies the effectiveness and stability of the spectral channel design, providing a reliable basis for subsequent spectral reconstruction and material classification experiments.

B. Spectral Reconstruction

To validate the spectral reconstruction capability of the proposed compressive hyperspectral imaging system, experiments were conducted on objects composed of Lego bricks with different colors. Fig. 5(a) shows the test scene, while Fig. 5(b) presents the grayscale measurement image captured under structured hyperspectral illumination. Regions of interest (ROIs) were selected from the measurement image for spectral analysis.

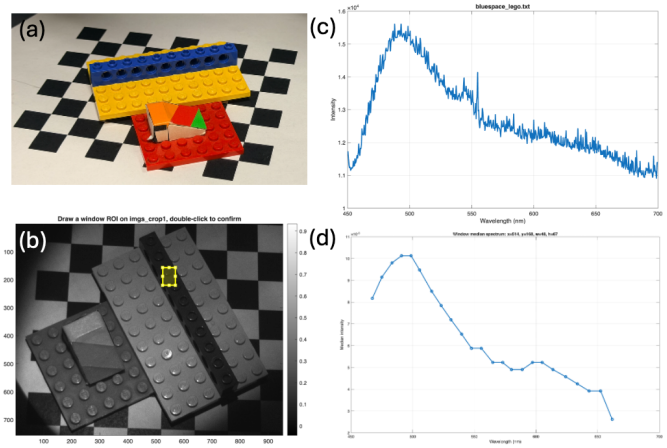

Fig. 5. Spectral reconstruction validation. (a) Photograph of the test object composed of Lego bricks with different colors. (b) Grayscale measurement image captured by the camera with the selected ROI marked. (c) Reference spectrum of the blue Lego measured using a spectrometer. (d) Reconstructed spectrum of the blue Lego obtained using the compressive hyperspectral imaging method.

The reference spectrum of the blue Lego region measured using a spectrometer is shown in Fig. 5(c). The spectrum reconstructed using the proposed compressive sensing–based hyperspectral imaging method is presented in Fig. 5(d). The reconstructed spectrum exhibits a similar overall shape and peak distribution compared with the reference measurement, demonstrating the effectiveness of the spectral encoding and reconstruction framework.

IV. Discussion

The results demonstrate that the proposed compressive hyperspectral imaging framework can achieve reliable spectral calibration and reconstruct material spectra with good agreement to spectrometer measurements, validating the effectiveness of spectral multiplexing at the illumination stage. By combining a hyperspectral projector, a grayscale camera, and computational decoding, the system provides a simple and potentially low-cost solution for accelerated hyperspectral imaging and real-time visual feedback. In future work, we will extend this framework to three-dimensional reconstruction and further implement it in the SWIR range, with the goal of achieving simultaneous spectral reconstruction and 3D imaging for surgical applications.

V. Acknowledgements

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VI. Credit Author Statement

Yunan Wang: Investigation, Writing – original draft, Data curation, Software. Lauren Gold: Conceptualization. Kevin F. Kelly: Conceptualization, Methodology, Resources, Supervision. Robert LiKamWa: Conceptualization, Methodology, Resources, Supervision.

REFERENCES

[1] S. Mironov, C. D. Hwang, J. Nemzek, J. Li, K. Ranganathan, J. T. Butts, D. J. Cholok, V. A. Dolgachev, S. C. Wang, M. Hemmila, P. S. Cederna, M. D. Morris, O. Berenfeld, and B. Levi, "Short-wave infrared light imaging measures tissue moisture and distinguishes superficial from deep burns," *Wound Repair and Regeneration*, vol. 28, no. 2, pp. 185–193, 2020.

[2] T. Du, D. K. Mishra, L. Shmuylovich, A. Yu, H. Hurbon, S. T. Wang, and M. Y. Berezin, "Hyperspectral imaging and characterization of allergic contact dermatitis in the short-wave infrared," *Journal of Biophotonics*, vol. 13, no. 9, p. e202000040, 2020.

[3] A. Bhargava, A. Sachdeva, K. Sharma, M. H. Alsharif, P. Uthansakul, and M. Uthansakul, "Hyperspectral imaging and its applications: A review," *Heliyon*, vol. 10, no. 12, p. e33208, 2024.

[4] S. Golovynskyi, I. Golovynska, O. Roganova, A. Golovynskyi, J. Qu, and T. Y. Ohulchanskyy, "Hyperspectral imaging of lipids in biological tissues using near-infrared and shortwave infrared transmission mode: A pilot study," *Journal of Biophotonics*, vol. 16, no. 7, p. e202300018, 2023.

[5] R. H. Wilson, K. P. Nadeau, F. B. Jaworski, B. J. Tromberg, and A. J. Durkin, "Review of short-wave infrared spectroscopy and imaging methods for biological tissue characterization," *Journal of Biomedical Optics*, vol. 20, no. 3, p. 030901, 2015.

[6] D. Roblyer, A. Pilvar, T. Pham, S. Spink, C. Hayakawa, V. Venugopalan, and M. Pierce, "Review of shortwave infrared imaging and spectroscopy in tissue [invited]," *Biomedical Optics Express*, vol. 16, pp. 5028–5062, 2025.

[7] R. H. Wilson, G. T. Kennedy, C. A. Campbell, T. Phan, A. H. Lin, B. Levi, and A. J. Durkin, "Short-wave infrared (swir) spectroscopy and imaging of biological tissues: a decade of advancements (2016–2025)," *Journal of Biomedical Optics*, vol. 31, no. 1, p. 010901, 2025.

[8] L. Gao, R. Kester, N. Hagen, and T. Tkaczyk, "Snapshot image mapping spectrometer (ims) with high sampling density for hyperspectral microscopy," *Optics Express*, vol. 18, pp. 14 330–14 344, 2010.

[9] M. M. Antony, C. S. S. Sandeep, and M. V. Matham, "Hyperspectral vision beyond 3d: A review," *Optics and Lasers in Engineering*, vol. 178, p. 108238, 2024.

[10] J. Stegemann, F. Gröniger, K. Neutsch, H. Li, B. S. Flavel, J. T. Metternich, L. Erpenbeck, P. B. Petersen, P. N. Hedde, and S. Kruss, "High-speed hyperspectral imaging for near infrared fluorescence and environmental monitoring," *Advanced Science*, vol. 12, no. 16, p. e2415238, 2025.

[11] T. Sun and K. Kelly, "Compressive sensing hyperspectral imager," in *Frontiers in Optics 2009/Laser Science XXV/Fall 2009 OSA Optics & Photonics Technical Digest*, ser. OSA Technical Digest (CD). Optica Publishing Group, 2009, p. CTuA5.

[12] Y. Xu, A. Giljum, and K. Kelly, "A hyperspectral projector for simultaneous 3d spatial and hyperspectral imaging via structured illumination," *Optics Express*, vol. 28, pp. 29 740–29 755, 2020.